SYMMETRIC ONE-DEPENDENT COLORINGS OF REGULAR TREES

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ABSTRACT. For every integer $d \geq 3$, we construct a proper coloring of the infinite d-regular tree with finitely many colors whose law is one-dependent and invariant under every tree automorphism and every permutation of the colors. The construction uses symmetric tensors in a real Hilbert space. Orthogonal projections force adjacent colors to differ, tensor summation gives consistent finite-dimensional laws, and a uniform estimate on rooted trees proves their positivity. An explicit, unoptimized bound is $q(d) = 2^{2d+16}d^2(2^d-1)9^{d-1}$ colors.

1. INTRODUCTION

A random proper coloring has local exclusions: neighboring vertices must receive different colors. Finite dependence asks for exact independence once two sets of vertices are sufficiently separated, despite those exclusions. Let T_d be the infinite connected tree in which every vertex has degree d, and let $[q] = \{1, \dots, q\}$. A field $X = (X_z)_{z \in V(T_d)}$ taking values in $[q]$ is *one-dependent* if $X|_A$ and $X|_B$ are independent whenever $A, B \subset V(T_d)$ are finite and their graph distance is greater than one. Here independence refers to the complete vectors, including their internal correlations. The field is proper if $X_z \neq X_w$ for every edge zw, almost surely.

Theorem 1.1. *For every integer $d \geq 3$, put*

$$q(d) = 2^{2d+16}d^2(2^d-1)9^{d-1}. \tag{1}$$

There is a proper one-dependent $[q(d)]$-valued coloring of T_d whose law is invariant under the full group $\mathrm{Aut}(T_d)$ and under every permutation of $[q(d)]$.

The number of colors is fixed before any finite set is considered. The bound in (1) is chosen for a uniform positivity estimate and is not claimed to be optimal.

Holroyd and Liggett constructed a stationary one-dependent four-coloring of $\mathbb{Z}$ with a law invariant under color permutations [1, Theorems 1 and 4]. Liggett and Tang studied one-dependent colorings of the three-ray star graph and asked about automorphism-invariant one-dependent colorings of regular trees [2]. Their star has a single branch point; a regular tree branches at every vertex. For unimodular transitive graphs of bounded degree, Timár obtained automorphism-invariant two-dependent colorings [3, Theorem 2]. Theorem 1.1 gives one-dependence on regular trees together with full color-permutation invariance.

Our construction is inspired by the word-space projections and positive-cone argument for paths in Flin, Li and Suomela [4, Section 4.1 and Appendix B]. We prove the estimates needed here directly. Branching requires a tensor with one slot for each incident edge, rather than a succession of path operators. Its sum over colors must allow deletion of an internal vertex, and positivity must survive simultaneous contraction against several child branches. An inverse sum of tensor projections supplies the deletion identity. A small neighborhood of a product tensor supplies positivity on every finite tree. The two parts of a tree bipartition use complementary projection ranges; conjugation by word parity removes the apparent dependence on that choice.

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Section 2 constructs the tensors and proves a quantitative approximation by product tensors. Section 3 obtains positive, consistent laws on finite forests. Section 4 extends them to the infinite tree and proves the required independence and symmetries. All Hilbert spaces and contractions below are real.

2. Projections on the word space

Fix $d \geq 3$. Initially let q be any integer satisfying

$$q \geq 32d16^d. \tag{2}$$

Let G consist of the empty word and all finite words in $[q]$ whose adjacent letters are distinct. On $H = \ell^2(G;\mathbb{R})$ write e_g for the coordinate unit vector at a word g, and write e for the unit vector at the empty word. Define L_i by prepending i to a word unless its first letter is i, in which case that letter is deleted. Define R_i by the analogous operation at the last letter. These operations permute the coordinate vectors.

Both operations are involutions. For a fixed i, the two end operations commute. On words of length at least two this follows by performing the end operations; on the empty word both composites return the empty word. On a one-letter word j, both composites return i if $j = i$, and iji otherwise. Thus L_i and R_i are commuting self-adjoint isometries. Set

$$P_i = \frac{(I + L_i)(I + R_i)}{4}, \qquad Je_g = (-1)^{|g|}e_g, \qquad Q_i = JP_iJ. \tag{3}$$

The operator J is a self-adjoint isometry, $Je = e$, and J anticommutes with L_i and R_i. Consequently

$$Q_i = \frac{(I - L_i)(I - R_i)}{4}, \qquad P_iQ_i = 0.$$

Both P_i and Q_i are orthogonal projections. The two vectors

$$u_i = e + e_i, \qquad v_i = e - e_i \tag{4}$$

satisfy

$$\|u_i\| = \|v_i\| = \sqrt{2}, \qquad \langle e, u_i\rangle = \langle e, v_i\rangle = 1, \qquad \langle u_i, v_j\rangle = \begin{cases} 0 & i = j, \\ 1 & i \neq j. \end{cases} \tag{5}$$

Here e_i is the coordinate vector at the one-letter word i.

For $k \geq 1$ we identify $H^{\otimes k}$ with $\ell^2(G^k;\mathbb{R})$, and set $H^{\otimes 0} = \mathbb{R}$. Tensor products are therefore ordinary products of coordinates. Contraction of a slot against $h \in H$ is a bounded map of norm at most $\|h\|$, by Cauchy–Schwarz. These conventions also define every partial contraction used below.

Lemma 2.1. *The bounded operator*

$$A = \sum_{i=1}^{q} P_i^{\otimes d}$$

on $H^{\otimes d}$ is invertible under (2), and

$$\|A^{-1}\| \leq \frac{2\,4^d}{q}. \tag{6}$$

Proof. Expand $P_i^{\otimes d}$ into its 4^d patterns, choosing in each coordinate one of I, L_i, R_i, L_iR_i. Each pattern defines an involutive permutation ϕ_i of G^d. Separate its fixed points from its off-diagonal entries. After averaging the patterns, write $A = D + B$, where D is diagonal and B has

zero diagonal. The all-identity pattern contributes $q/4^d$ to every diagonal entry, whereas each summand of A has diagonal entries at most one. Thus

$$\frac{q}{4^d} \le D(g,g) \le q. \tag{7}$$

For a fixed pattern, orient each nontrivial edge $g \leftrightarrow \phi_i(g)$ from smaller to larger total word length, using a fixed total order to break ties. At any tuple, at most $2d$ labels i can give a nontrivial neighbor of no larger length. Indeed, such an i must be the first or last letter of a nonempty coordinate on which the pattern acts. Outside this set of labels, a single end operation increases the length by one. The operation L_iR_i increases the length of a nonempty word by two, and fixes the empty word. If every active coordinate is empty and every active operation is L_iR_i, the tuple is fixed, so it was already removed.

Let M be the matrix of the oriented edges, retaining multiplicities if different labels give the same edge. One of its row and column sums is at most q, and the other is at most $2d$. For a finitely supported f, weighted Cauchy–Schwarz gives

$$\|Mf\|_2^2 \le \left(\sup_x \sum_y M_{xy}\right)\left(\sup_y \sum_x M_{xy}\right)\|f\|_2^2 \le 2dq\|f\|_2^2.$$

It extends to $H^{\otimes d}$ by density. The off-diagonal matrix of this pattern is $M + M^*$, so averaging all patterns proves

$$\|B\| \le 2\sqrt{2dq}. \tag{8}$$

Equations (7)–(8) imply $\|D^{-1}B\| \le 2\,4^d\sqrt{2d/q} \le 1/2$. The norm-convergent Neumann series for $(I + D^{-1}B)^{-1}$ now gives (6) from $A = D(I + D^{-1}B)$. □

Define

$$V = A^{-1}e^{\otimes d}, \qquad t_{i,d} = P_i^{\otimes d}V. \tag{9}$$

For $0 \le k \le d$, obtain $t_{i,k}$ by contracting $d-k$ slots against e, and set $s_{i,k} = J^{\otimes k}t_{i,k}$. The operators A and A^{-1} commute with permutations of the d slots. Hence V and $t_{i,d}$ are symmetric tensors, and the definition of $t_{i,k}$ does not depend on which slots are retained. The tensors satisfy

$$\sum_i t_{i,k} = \sum_i s_{i,k} = e^{\otimes k}, \tag{10}$$

$$t_{i,k} \in \operatorname{ran}(P_i^{\otimes k}), \qquad s_{i,k} \in \operatorname{ran}(Q_i^{\otimes k}). \tag{11}$$

Contracting any slot of $t_{i,k}$ against e gives $t_{i,k-1}$, and the same holds for $s_{i,k}$ because $Je = e$. A permutation π of $[q]$ acts on every letter of G, fixes e, and conjugates P_i to $P_{\pi(i)}$. It commutes with A after summation. Thus it carries $t_{i,k}$ and $s_{i,k}$ to the tensors indexed by $\pi(i)$.

Lemma 2.2. *Put $C_d = 2\,4^d\sqrt{2^d-1}$ and $\delta = C_dq^{-3/2}$. For every i and $0 \le k \le d$,*

$$\|t_{i,k} - q^{-1}u_i^{\otimes k}\| \le \delta, \qquad \|s_{i,k} - q^{-1}v_i^{\otimes k}\| \le \delta. \tag{12}$$

Proof. Let $V_* = 2^de^{\otimes d}/q$. Since $P_ie = u_i/2$,

$$AV_* - e^{\otimes d} = \frac{1}{q}\sum_{i=1}^{q}\sum_{\varnothing\neq S\subseteq[d]}\bigotimes_{j=1}^{d} h_j(i,S), \qquad h_j(i,S) = \begin{cases} e_i & j \in S, \\ e & j \notin S. \end{cases}$$

These $q(2^d-1)$ terms are distinct orthogonal coordinate vectors. The residual therefore has norm $\sqrt{(2^d-1)/q}$. Lemma 2.1 gives $\|V - V_*\| \le C_dq^{-3/2}$. Apply $P_i^{\otimes d}$ and contract the discarded

slots against e, operations of norm at most one. The corresponding contraction of $P_i^{\otimes d} V_*$ is $q^{-1} u_i^{\otimes k}$. Applying the isometry $J^{\otimes k}$ proves the second estimate. □

3. Probabilities on finite forests

Choose either bipartition of T_d. For a finite vertex set F and a specified coloring $x \in [q]^F$, use its induced forest. At a vertex z of degree k in that forest, place $t_{x_z,k}$ if z is in the first part and $s_{x_z,k}$ if it is in the second. Assign the tensor slots to incident edges in any order and contract the slots at the two ends of each edge by the real inner product. Let $p_F(x)$ be the product of the resulting scalars over components. An isolated vertex uses a tensor with $k = 0$, and $p_\varnothing = 1$.

These finite-tree contractions are well-defined. Contracting a leaf vector into its neighbor has norm at most the product of the two norms. Induction bounds a full scalar contraction in absolute value by the product of all vertex-tensor norms, and bounds an outgoing vector contraction in the same way. The contraction is multilinear and continuous in the finitely many tensors. Approximate them by finite-support tensors: their finite sums agree in all leaf-contraction orders, and continuity gives order independence for the original tensors. No infinite tree is contracted in this definition.

Proposition 3.1. *The functions p_F vanish on improper colorings and satisfy, for each $z \in F$,*

$$\sum_{a=1}^{q} p_F(x_{F\setminus\{z\}}, x_z = a) = p_{F\setminus\{z\}}(x_{F\setminus\{z\}}). \tag{13}$$

They are invariant under color permutations and under tree automorphisms. In particular, $\sum_{x\in[q]^F} p_F(x) = 1$.

Proof. Cut a monochromatic edge with color i at both ends and contract the branches on its two sides. By (11), the remaining endpoint vectors lie in the orthogonal ranges of P_i and Q_i. Their inner product is zero.

To sum over a vertex color, (10) replaces its tensor by $e^{\otimes k}$. Each of its incident edges then contracts e into the tensor at the other endpoint, producing the prescribed tensor with one fewer slot. This proves (13), including when deletion of an internal vertex splits a component into several branches. Repeated deletion proves the total mass identity and consistency under every restriction of finite sets.

The action of a color permutation on the word space preserves inner products and permutes the tensor family, so it preserves p_F. Switching the two parts of the bipartition applies J at both ends of every edge. Since J is an isometry, those two actions leave each inner product unchanged. Thus the law is independent of the bipartition choice. Slot symmetry removes the choice of incident-edge order. Every automorphism of the connected tree preserves or swaps its two bipartition classes, proving the stated tree invariance. □

It remains to prove that the scalars $p_F(x)$ are nonnegative. Orthogonality and summation alone do not give this. We now take $q = q(d)$ from (1); it satisfies (2). With the constant C_d of Lemma 2.2, this choice can be written as

$$q = \left[64 d C_d (3/2)^{d-1}\right]^2, \qquad \eta = \frac{1}{16d}, \qquad \tau = q\delta(3/2)^{d-1} = \frac{1}{64d} = \frac{\eta}{4}. \tag{14}$$

Lemma 3.2. *For this value of q, every proper coloring of every finite induced forest has $p_F(x) > 0$.*

Proof. First consider a finite rooted branch with one additional outgoing edge at its root. Use one tensor slot for that edge, and contract all edges within the branch, obtaining a message

$m \in H$. We prove by induction that, for a properly colored branch with root color i in the first part,

$$m = z\ell, \qquad z = \langle e, m\rangle > 0, \qquad \langle e, \ell\rangle = 1, \qquad \|\ell - u_i\| \le \eta. \tag{15}$$

For a root in the second part replace u_i by v_i. The message remains in the corresponding projection range by (11).

Suppose the root has $r \le d-1$ children. By induction write their messages as $z_j\ell_j$ with $z_j > 0$; the case $r = 0$ uses empty products and requires no induction hypothesis. A root in the first part has children in the second part, with colors $i_j \ne i$. After dividing out these already positive scalars, Lemma 2.2 gives

$$\frac{m}{\prod_j z_j} = cu_i + \epsilon, \qquad c = \frac{1}{q}\prod_{j=1}^{r}\langle u_i, \ell_j\rangle, \qquad \|\epsilon\| \le \delta(3/2)^r. \tag{16}$$

Indeed, $\|\ell_j\| \le \sqrt{2} + \eta < 3/2$. From $\langle u_i, v_{i_j}\rangle = 1$ and (15), each factor in qc is at least $1 - \sqrt{2}\eta$. Therefore

$$qc \ge (1 - \sqrt{2}\eta)^r \ge 1 - r\sqrt{2}\eta \ge \frac{7}{8}. \tag{17}$$

The scalar $b = \langle e, cu_i + \epsilon\rangle$ is at least $(7/8 - \tau)/q > 0$. Normalizing (16) by b yields

$$\left\|\frac{cu_i + \epsilon}{b} - u_i\right\| = \left\|\frac{\epsilon - \langle e, \epsilon\rangle\, u_i}{b}\right\| \le \frac{(1+\sqrt{2})\tau}{7/8 - \tau} \le 4\tau = \eta.$$

Thus (15) holds at the root, including the leaf case. At a root in the second part, the same calculation uses v_i and the overlaps $\langle v_i, u_{i_j}\rangle = 1$.

Finally root a whole tree component, with no outgoing edge. The root can have $r \le d$ children. Its scalar contraction is at least

$$\frac{\prod_j z_j}{q}\left[\frac{7}{8} - q\delta(3/2)^d\right] > 0. \tag{18}$$

Here (17) still holds because $r \le d$ and $d\sqrt{2}\eta = \sqrt{2}/16 < 1/8$, while $q\delta(3/2)^d = (3/2)\tau \le 3/128$. This also covers an isolated vertex. The product over components is strictly positive. The same q and cone radius work for every finite size. □

4. The infinite coloring

Proof of Theorem 1.1. Proposition 3.1 and Lemma 3.2 show that p_F is a probability mass function on $[q]^F$ for every finite vertex set F, and that these functions are projective under restriction. The vertex set of T_d is countable, and the discrete state space $[q]$ is a Borel space. The extension theorem [5, Theorem 5.16, p. 92] therefore gives a field X having exactly these finite-dimensional laws.

If finite sets A, B have distance greater than one, their induced union has no edge between A and B. Every component belongs entirely to one of the two sets. The definition by component contractions gives, for all $x \in [q]^A$ and $y \in [q]^B$,

$$\mathbb{P}(X|_A = x, X|_B = y) = p_{A\cup B}(x, y) = p_A(x)p_B(y).$$

This is independence of the complete vectors. Empty restrictions are deterministic and cause no exception.

All finite-dimensional laws have the two invariances proved in Proposition 3.1. Finite-dimensional distributions determine the law on the product measurable space [5, Proposition 2.2, p. 25]. Applying this to each tree automorphism and each color permutation gives invariance of the entire law. The transformations are measurable since each transformed coordinate is a

finite-state function of one original coordinate. For every edge, the probability of equal endpoint colors is zero. There are countably many edges, so all edges are proper simultaneously almost surely. □

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